

Project Initialization and Planning Phase

Date	01 July 2026
Team ID	xxxxxxx
Project Title	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **OptiCrop – AI-Based Crop Recommendation System** is designed to recommend the most suitable crop for cultivation using machine learning techniques. The system analyzes agricultural parameters such as **soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, soil pH, and rainfall** to generate accurate crop recommendations.

Traditional crop selection depends on experience and manual analysis, which may lead to incorrect decisions and reduced productivity. OptiCrop automates this process by using a trained machine learning model to provide quick and reliable recommendations. The web application enables farmers to enter field details and instantly receive the best crop suggestion, helping improve productivity and resource utilization.

Key features include **input validation, data preprocessing, machine learning prediction, crop recommendation, and result visualization** through a simple web interface.

Project Overview	
Objective	The primary objective of this project is to develop an AI-based crop recommendation system that predicts the most suitable crop based on soil and environmental conditions. The system assists farmers in making informed agricultural decisions, improving crop yield, and reducing the risk of selecting unsuitable crops.
Scope	The project focuses on collecting agricultural datasets, training a machine learning model, predicting suitable crops based on user inputs, and presenting recommendations through a user-friendly web application. The

	solution supports farmers, agricultural researchers, and students by providing fast and accurate crop recommendations.
Problem Statement	
Description	Farmers often face difficulties in selecting the most suitable crop because soil properties and environmental conditions vary across locations. Manual decision-making is time-consuming and may result in poor crop selection, reduced productivity, and financial loss.
Impact	Incorrect crop selection can lead to low agricultural yield, inefficient use of fertilizers and water, and economic losses. An AI-based crop recommendation system improves decision-making by providing accurate and data-driven crop suggestions.
Proposed Solution	
Approach	Develop an AI-powered crop recommendation system using machine learning that analyzes Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall to recommend the most suitable crop for cultivation through a Flask-based web application.
Key Features	• Data preprocessing and cleaning • User input validation • Machine learning model training • Crop recommendation prediction • Fast web-based prediction interface • Accurate recommendation results • Simple and responsive user interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i5 or higher
Memory	RAM specifications	8 GB
Storage	Disk space	20 GB Free SSD/HDD

Software		
Operating System	Platform	Windows 10/11 or Ubuntu
Programming Language	Development	Python 3.10+
Framework	Web Framework	Flask
Libraries	Machine Learning	Pandas, NumPy, Scikit-learn, Matplotlib, Joblib
IDE	Development Environment	VS Code
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Crop Recommendation Dataset (CSV) containing agricultural parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall, used to train and evaluate the machine learning model for accurate crop recommendations. Source: Kaggle Crop Recommendation Dataset.